Math 344, F18 **Name:** Answer

Quiz 10

__Instructions:__ **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (6 points) Find an orthonormal basis $\mathbf{q}_1$, $\mathbf{q}_2$, $\mathbf{q}_3$ for the column space of $\mathbf{A} = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}$ using the *Gram-Schmidt* process.

Solution: 1. $\tilde{q}_1 = a_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\|\tilde{q}_1\| = \sqrt{1^2+0+0} = 1$

$$\Rightarrow q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$a_2^T q_1 = 2\cdot 1 + 0\cdot 0 + 3\cdot 0 = 2$

2. $\tilde{q}_2 = a_2 - (a_2^T q_1) q_1 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - (2)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$= \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

$\|\tilde{q}_2\| = \sqrt{0^2+0^2+3^2} = 3 \Rightarrow q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|} = \frac{1}{3}\begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

3. $\tilde{q}_3 = a_3 - (a_3^T q_1) q_1 - (a_3^T q_2) q_2$ $\quad a_3^T q_2 = 4\cdot 0 + 5\cdot 0 + 6\cdot 1 = 6$

$a_3^T q_1 = 4\cdot 1 + 5\cdot 0 + 6\cdot 0 = 4$

$$= \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - (4)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 6\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$$

$\|\tilde{q}_3\| = \sqrt{0^2+5^2+0^2} = 5 \Rightarrow q_3 = \frac{\tilde{q}_3}{\|\tilde{q}_3\|} = \frac{1}{5}\begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

Thus $Q = [q_1\ q_2\ q_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

Problem 2. (4 points) True or false.

(a) The determinant of $\mathbf{A} - \mathbf{B}$ equals $\det\mathbf{A} - \det\mathbf{B}$.

False

$$\det\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \det\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 4 - 1 = 3 \neq \det\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\downarrow$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(b) $\mathbf{AB}$ and $\mathbf{BA}$ have the same determinant.

True

$$\det(AB) = \det A \cdot \det B = \det(BA)$$

(c) $\det\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = 1$

True

$$\det\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = -\det\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \det\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 1$$

(d) $\det(4\mathbf{A}) = 4\det(\mathbf{A})$.

False

$$\det(4A) = 4^n \det(A)$$